

Random Variables

Suppose you are a quality manager at a chips manufacturing plant. The quality of a batch is checked by picking 3 bags at random and checking the quality of each one. The sample space for this experiment can be expressed as:

$$S = PPP, PPF, PFP, FPP, FFP, FPF, PFF, FFF$$

As the quality manager, one natural concern would be: “how many bags fail the quality test?” To each element in the sample space, we can assign a number that represents the number of failed bags:

A random variable (RV) is a real – valued function defined over a sample space. A RV assigns a value to each outcome in the sample space. Typically, capital letters at the END of the alphabet are used to denote a RV. The corresponding lower case letter is then used to denote a specific value.

The probability that X takes on the value x , $P(X = x)$ is defined as the sum of the probabilities of all sample points in S that are assigned the value x . We will sometimes denote $P(X = x)$ by $f(x)$.

Using the previous example, we can find $P(X = 0)$, $P(X = 1)$, $P(X = 2)$, and $P(X = 3)$.

Discrete Random Variables

A discrete random variable can assume only a finite or countably infinite number of values (counting numbers). Most often interest lies in how the probabilities are distributed over all possible values of the sample space. This probability distribution can be given in the form of a table, graph, or formula.

Example: workers consist of 3 men and 3 women. Suppose 2 workers are chosen at random. Note wanting to show bias, the foreman is interested in $X =$ number of women selected. Find the probability distribution of X .

- What are the possible values of X ?
- How many elements are in S ?
- What are the probabilities of each outcome in X occurring?
- Table

- Graph

- Formula

A probability mass function (pmf) is a set of ordered pairs $(x, f(x))$ of the discrete random variable X if, for all possible values of x :

- $P(X = x) = f(x)$
- $0 \leq f(x) \leq 1$
- $\sum_x f(x) = 1$

Example: find the value of c so that $f(x)$ is a pmf for a random variable X .

$$f(x) = \frac{x}{c} \text{ where } x = 1, 2, 3, 4$$

Example: is $f(x)$ a valid pmf?

$$f(x) = 0.3^x \times 0.7^{1-x}; \quad x = 0, 1$$

The cumulative distribution function (cdf), $F(x)$ of a discrete random variable with pmf $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t); \quad -\infty \leq x \leq \infty$$

Example: let $X = 1, 2, 3, 4$ and $f(1) = 0.1$, $f(2) = 0.3$, $f(3) = 0.25$, $f(4) = 0.35$. Create a table and graph for $F(x)$.

The **expectation** of a random variable X is a weighted average, denoted $E[X]$, of the possible

values that X can assume.

Formal Definition: If X is a discrete random variable having probability mass function $f(x)$, then the expectation (expected value or mean) of X , denoted $E[X]$, is defined by:

$$\mu_x = E[X] = \sum_{f(x)>0} x \cdot f(x)$$

Example: find the expected value of the pmf $f(x) = \frac{1}{30}(x+1)^2$ where $x = 0, 1, 2, 3$

Proposition If X is a discrete random variable that takes on the values of x_i where $i \geq 1$, with respective probabilities $f(x_i)$, for any real-valued function g ,

$$E[g(X)] = \sum g(x_i)f(x_i)$$

Example: find $E[X^2]$ of the pmf $f(x) = \frac{1}{30}(x+1)^2$ where $x = 0, 1, 2, 3$

Corollary: If a and b are constants, then

$$E[aX + b] = a \cdot E[X] + b$$

Let $E[X]$ be 5 and $E[X^2]$ be 15 for a random variable X . Find the following:

- $E[x - 1]$

- $E[3x]$

- $E[-2x + 5]$

Another characteristic of interest for a discrete random variable is the variance which will provide an indication of variation or spread of the values of X .

Definition: If X is a random variable with mean, μ , then the variance of X , denoted by $Var(X)$, is defined by

$$Var(X) = \sigma_X^2 = E[(X - \mu)^2] = E[X^2] - (E[X])^2$$

Proof:

Example: find the variance for the pmf $f(x) = \frac{x}{10}$, where $x = 1, 2, 3, 4$

Corollary: for any constants a and b , then

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Proof:

Let $E[X] = 5$ and $E[X^2] = 15$. Find the following:

- $Var(X)$

- $Var(3X + 5)$